



SRI SRINIVASA EDUCATIONAL & CHARITABLE TRUST (R)
SAPTHAGIRI COLLEGE OF ENGINEERING
(Affiliated to Visvesvaraya Technological University, Belagavi and Approved by AICTE-New Delhi)
(Accredited by NAAC "A" Grade & NBA) (ISO 9001:2015 & ISO 14001:2015 Certified)
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DEPARTMENT OF MATHEMATICS
ACADEMIC YEAR: 2023-24 EVEN SEMESTER (BMATS201)
QUESTION BANK

MODULE-3: VECTOR SPACE AND LINEAR TRANSFORMATION

1. Prove that the subset $W = \{(x, y, z) : x - 3y + 4z = 0\}$ of the vector space R^3 is a subspace of R^3 .
2. Let $V = R^3$ be a vector space and consider the subset W of V consisting of vectors of the form (a, a^2, b) , where the second component is the Square of the first. Is W a subspace of V ?
3. Prove that in $V_3(R)$, the vectors $\{(1, 2, 1), (2, 1, 0), (1, -1, 2)\}$ are linearly independent.
4. Show that the set $S = \{(1, 2, 4), (1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is linearly dependent.
5. Define linearly independent vectors and linearly dependent vectors. Are the vectors $V_1 = (2, 5, 3)$, $V_2 = (1, 1, 1)$ and $V_3 = (4, -2, 0)$ are linearly independent? Justify your answer.
6. Express the vector $(3, 5, 2)$ as a linear combination of the vectors $(1, 1, 0)$, $(2, 3, 0)$, $(0, 0, 1)$ of $V_3(R)$.
7. Determine whether the matrix $\begin{bmatrix} -1 & 7 \\ 8 & -1 \end{bmatrix}$ is a linear combination of $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $\begin{bmatrix} 2 & -3 \\ 0 & 2 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$ in the vector space $M_{2 \times 2}$ of 2×2 matrices.
8. Define a basis for a vector space. Determine whether or not the vectors $(1, 1, 2)$, $(1, 2, 5)$, $(5, 3, 4)$ form a basis of R^3 .
9. Find the dimension and basis of the subspace spanned by the vectors $(2, 4, 2)$, $(1, -1, 0)$, $(1, 2, 1)$ and $(0, 3, 1)$ in $V_3(R)$.
10. Prove that $T: R^3 \rightarrow R^3$ be defined by $T(a, b, c) = (3a, a - b, 2a + b + c)$ is a linear transformation.
11. Prove that the transformation $T: R^2 \rightarrow R^2$ defined by $T(x, y) = (3x, x + y)$ is linear. Find the image of the vectors $(1, 3)$ and $(-1, 2)$ under this transformation.
12. Prove that $T: R^3 \rightarrow R^3$ be defined by $T(a, b, c) = (3a, a - b, 2a + b + c)$ is a linear transformation.
13. Find the matrix of linear transformation $T: V_2(R) \rightarrow V_3(R)$. Such that $T(-1, 1) = (-1, 0, 2)$ and $T(2, 1) = (1, 2, 1)$.
14. the matrix of linear transformation $T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x + y, y + z)$ w.r.t the basis $\{(1, 1, 1), (1, 0, 0), (1, 1, 0)\}$ and also w.r.t the standard basis.
15. Find the linear transformation $T: R^2 \rightarrow R^2$ satisfying $T(1, 1) = (0, 1)$ and $T(-1, 1) = (3, 2)$

16. Verify Rank nullity theorem for the linear transformation $T:R^3 \rightarrow R^2$ defined by $T(x,y,z)=(y-x, y-z)$
17. State rank nullity theorem and verify the theorem for the linear transformation $T:R^3 \rightarrow R^3$ defined by $T(x,y,z)=(x+2y-z, y+z, x+y-2z)$.
18. Let $T:V \rightarrow W$ be a linear transformation defined by $T(x,y,z)=(x+y, x-y, 2x+z)$. Find the range, null space, rank, nullity and hence verify the rank-nullity theorem.
19. Find the kernel and range of the linear operator $T(x,y,z)=(x+y, z)$ of $T:R^3 \rightarrow R^2$.
20. Show that the function $f(x)=3x-2$ and $g(x)=x$ are orthogonal in P_n with inner product $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$.